

Name	Origin & distribution	Soil & climate	Varieties	Sowing time & spacing	Field preparation	Propagation method / seed rate & treatment	Fertilizer Water management	Training & pruning	Insect & control	Disease & control	Harvesting	yield
1. ROSE (<i>Rosa indica</i> L.) <i>Rosaceae</i>	National f. of England Alaska Mexico D.-france Greece, India Iran, USA for perfume	Sandy loam pH- 6-7.5 & n. temp 15-18. & day t. 20-25	Rosa cinensis, ro. Fhitida, galika, gigentia, masketa, multiflora, Wild spp.-clinofilla, serisia, macrophylla	Sep-oct. Feb-mar. 90-100cm.	Plough the land 4-5 times 15 day dry in the sun Pit size- 45x45x30cm. Soil manure mixture – soil 40%, cow dung 40%, leaf mold 10% bone meal 10% Urea 25g, MOP 100g SSP 50g.	Cutting – 15-20cm length. & 3-4 nodes cutting are treated with IAA, BA 100ppm Budding – T budding . Layering and gootee. By seed keep. 1-2 hr. sulfuric acid for fast germ.	6x6x12 npk - 8x8x12 /plant & 500x850x700/hq. 30x40cm par 8-10 day int. flood irri. In summer 5-6 day	Old stem pruned after winter 3-4 month 1 st pruned after planting Cut plant x 45 degree angle	Aphid, thrips jassid. Termite, ash weevil, leaf rolling larva Ctr.- malathion, metasystox.	Powdery mildew, black spot, rust, die back Ctr, karathane, mineb	Jul-aug Plant on yield on economic Scale for 3 years.	225-350 flower /sq m.
2. GERBER (<i>Gerbera jamesonii</i>) <i>compositae</i>		Slightly alkaline soil pH 5.5-7.0 c. reqir. Shed polyhouse day t. 22-25 night t. 12-16	red- besta beauty, dusty, ylw – doni, supernova mamut, talas. Pnk- pink Elegance, marmara, esmara Org- Carrera, gliasth, marsol.	the crop can be cultivated throughout of the year spc-40x30cm /30x30cm	1 cultivation mb plough then 2-3 culti. After / cultivation use PATA	Division of suckers and tissue culture plants	7.5 kg /sq m cow dung or compost Drip irri. Avg. Wtr. Req. is 500-700 ml/day/plant	leaf pruning remove old leaf periodically	Aphids, whitefly, thrips, nematode Ctr. Imidacloprid, fipronil, abamectine	. Flower bud rot, powdery mildew. wilt. Stem break. Ctr. Copper oxychloride. Wettable sulphur.	.3 rd month of planting, and conti. Of 2 years	.2 ^{ste} m/ plant /month
3. carnations (<i>Dianthus caryophyllus</i>) <i>caryophyllaceae</i>		Red loam pH, 5.5-6.5 Tempret (cool) Tem. Day 18-24 ni. 10-15	Red-domingo, big red. white- white liberli, white dona. lisa, pink – dona,	15x15 (22-25 plant/ m sq)		suckers, terminal cut 5-10cm. trt. NAA 500ppm of 5 minuts carbendazim	Neem cake - 2.5 tan/ha Pho-400g/100sq ft Mg sulf. - 0.5kg/100sq ft Drip irri.		.Aphid, Thrips mites bud borer. Ctr.- abamectin, Bassila thuringiensis	Wilt, leaf spot. Crt. Carbendazim, pseudomonas	.flowering starts 110-120 days after planting	15 flower /plant. 350-375 flwr/ msqr. in 2 year
4. tuberose (<i>polianthes tuberosa</i> L.) <i>Amaryllidaceae</i>		Sandy loam pH, 6.5-7.5 Tem 20-35	Rajat rekha, svarna rekha Suvasini, Calcutta single maxican single, pune single, haidrabad single, phule rajni.	Mar-apr. 20x20cm	30-40 cm plough during jan.	Bulb.	20-25 tan FYM after 200:200:200 kg /ha. ploughing in soil. After planting. Summer time – april – jun. Weekly interval Winter 10 day interval		.thrips, aphids, nematode, Ctr. dimethoate. malathion, indosalphan.	.stem rot. Wilt Ctr. Carbendazim. Zineb. 0.3 %	.loos flower before. 8 am Cut flower. Spike is cut leaving 4-6 cm	.loos flower . 14-15t/ha Cut flr. 2-3 laks spike s /year.

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5.lilium (<i>Lilium spp.</i>) Liliaceae		.pH 5.5-6.5, DT-18-22, NT-10-15.	W-Novona , Y- Dreamland,Giant. Pi-vivaldi R- Black out	20x15cm, 15x15cm depending on bulb size		Bulb, 6 week cold storage period at 2dee. C to 5 dee.c is needed to break dormancy.	6-8 litters / m2/day in summer & witer 5-6 liters		Aphids,mites, thrips, Ctr. Dimethoate Abamectine, propargite.		Bulb&sca le rot. Foot&root rot.ctr. zineb , carbenda zim.	Bud show color then harvest. Color breakin g stage.	30-40 f / stem M2
6. Gladiolus (<i>gladiolus spp.</i>) <i>Iridacdeae.</i>		Sandy lomy pH, 6-7. Temprate./sub tropical. Tem.27-30	. Basant bahar. Indrani. Kalmia. Kohra, aarti. Arka kesar, darsan	.plain oct. Hill mar-apr. 40x25cm /25x28p/hac	.5cm. deep plough.	.corm. & there storage of 3-7cg. For 3 month	.NPK-60:150:150. 7-10 Day interval sd lm. soil		.thrips, semi looper helicoverpa, Ctr. Methyle demeton. Dimethoate. Melathion.		. leaf spot. Blight. Spot. Ctr. Benomyl. Carvenda zim. Mercuric.	.110-120 days	.2-2.5 lakh flower /hac/c rop
7.Chrysanthemum (<i>Chrysanthemum indicum</i>) <i>compositae</i>		Red lomy soil pH 6-7 Tropical & sub T. 20-28 tem. D& N-15-20	CO1,CO2,MDU1, Indira &red gold	Apr-may 30x30cm		Terminal cuttings 5-7cm suckers planting during jun-jul 111000 P/ha	25 tan FYM, 125x125x 25 NPK Irr. Weekly intervals		Thrips aphids & leaf eating caterpillars. Ctr. Acetamiprid , indoxacrb		Root rot, leaf spot mosaic disease Ctr. Moncozeb	Start from 3 rd month at 4 days interval	20qt/h ac
8.Mrigold (<i>Tagetes erecta.</i>) <i>compositae</i>		Lomy soil pH7-7.5 saline and acidic soil	Pusa narangi genda , pusa basanti genda MDU1	45x35cm		Nursery 1.5kg seed/ha transplanted after 4 weeks	25tan FYM, 45:90:75 N:P:K, &45 kg N/ha top dressing after 45 days planting Irr. Once in a week.		Mealy bug thrips, spider Ctr. Acetamiprid , indoxacrb		Leafspot root rot ctr. Bavistine	Afr 60 day of planting picked up once in 3 days	18 tan/ha
9.jasmin (<i>jasminum spp.</i>)		Sandy& rad lome tropical crop low rainfall areas its req. modrate humidity	H yild V- co-1 mullai, &2, Long round, short round , short point	Jun-jyl, oct-nov 1.2x1.2m, 1.5x1.5 m.		Cutting & layers	10-12 kg FYM/plant j. sambac 90:120:240for time intervals & others 60:120:120 Inrrigation twice in a week	Wthe pruning is done after 2 years of its planting and its followed in the month of dec-jan. feb-march	Bud &shoot borer, thrips stick bugs ctr.- dimethoate, carbyl, malathion		Leaf blight& rust Ctr.- bordeaux mixture, zineb	2 nd year starting	1-2 tan /ac
10.Aswagandha. (<i>withania somnifera</i>) <i>solanaceae</i>		EC/P- Root leaf	Jammu,laknow,j awahar 20	Rainy season aug-sep, 60x60cm/ nursery spa. At 5cm		nursery 5kg seed 10-12kg/ha	Hence no, manure of fertilizers are recomamnded.				Leaf blight & die back seedling rot Ctr. Diethane.	150-170days after sowing,	400-1200kg/ ha roots 200-500kgs eedh

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11. Asparagus (<i>asparagus officinalis</i>) <i>asparagaceae</i>	Europe & asia	Friable, good drainage pH-6-6.5 cl. temperate & tropics		July-nov. 1.5-2m & 45x60cm		Seed. Seed & planting both single & double rows	Irri.- every 2 weeks 50mm		Cutworms, asparagus beetle Ctr. thiodan	Wilt, rust, Ctr.	When tips of above the ridges	
12. Aloe (<i>aloe barbadensis mill</i>) Liliaceae		Lomy soil pH-7-8.5, tem. 25-45 deg. C		Jun -july & sep-oct 90x90cm		Suckers 10000/ha Tre. Dipped in 0.1% carbendazim 5 min then planting	10 tan FYM are req. 30kg N & 120kg gypsum. It's a rainfed crop. During summer 4-5 required.		No pest no disease		6-7 month gel extraction within 6 hours	15 tan /ha
13. COSTUS (<i>Zingiberaceae</i>)	Indo malyan region	Sandy clay lome pH 5.5-7.5, 1000-1500mm annual rainfall		50x50cm		Rhizome 40g should be selected 2000-2400kg/ha	15 ta/ha. FYM 45:30:30, NPK Irri. 2-3 times in a month			Rhizome rot, leaf blight Ctr. dithian, bavistin	16-17 month old plants can be harvested	28-30 t/ha rhizomes
14. cinnamomum (<i>cinnamomum verum</i>) lauraceae	Shrilanka Dis.-india, malabar	Sandy lome Red dark brown soil. 150-250cm rainfall. Temp 27 deg. C.		3x3 m		Root, suckers, & terminal cutting.						
15. Mint. (<i>mentha spp.</i>) labiateae	P	Lomy soil pH 6.5-7 Sub tropical 100-150cm rainfall	MAS-1, MAS-2, MSS-1 & 5, HYBRID-77	June-july 40x40cm		Suckers terminal cutting .	50:75:50kg/ha At four months AP N25 kg /ha is applied & irri. Is given twice in a week.				Frist - 5 month. And subsequent 3 months interval. 4 year	24-30 T of herbage & 250-300kg oil
16. periwinkle (<i>catharanthus roseus</i>) apocynaceae		Suit all soil Tropical climate. Rainfall 100 cm	White flowered, Purple flowered, And normal & Dhawal.	45-60 day after trans. 45x20cm Jun july or Sep-oct		2.5kg/ha seed.	20 t/ha FYM. 25:50:75kg/ha				Harvest of roots after 1 year.	7.5-15 qt 10-15 qt 20-30 qt..
16. occimum <i>occimum spp</i>		Hardy lome and sandy lome		4.5x1.0x2		300kg hectare	50:10:100 in applying to 30 day			Throw the seed	6-7 month after harvesting	

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17. geranium pelargonaceae belonging on family		A deep well drained and required to 5.5-6.0 pH and it is highly growth on 1000-1500 cm annual rainfall		110x60 cm		Harvesting at after on 120 day	50:30:50 N.P.K required during on September october				Oil yield 100 kg and cutting yield 250kg hectare	
18. palmarosa. (cymbopogon martinii) poaceae		Lomy soil tropical 150 rainfl. 9.5pH.	Motia, sofia, tripta, trisna, OPD-1,2, PRC-1,	Jun-july 60X60cm.		Seed & sleeps 2.5kg & 28000/ha.	10tan FYM 20:50:40/ha				3-4 month of interval For 5 year	Herbage. 20-30 t/ha Oil- 20 60. 70kg
19. vetiver (vetiveria zizanoides) poaceae		Sandy lome. pH 6- 8 100-200 cm rainfall.	Hybrid 8. Sugandha, ODV-3	Jun-july		Slips 150000-225000 slips/ha	60kg N 22.5 P. 10t FYM.		Termites and white ants, attack the roots.	Leaf blight Bordeaux mixture for ctrl.	18 month of planting	3-4 tones of roots . 15-16 kg oil.

AHFL-5221 BY PINKU
THANKS

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